

Assignment Sample

Course: B.Tech | Stream: Computer Science | Subject: Data Structures | Semester: III | Paper Code: DS301

Unit 2

Topics: Linked Lists; Stacks; Queues; Applications of Linear Data Structures

- Q1. Define a linked list and explain its advantages over an array. [2 Marks, L1]
- Q2. Differentiate between singly linked lists and doubly linked lists with examples. [3 Marks, L2]
- Q3. Write an algorithm to push and pop elements from a stack. [5 Marks, L3]
- Q4. Compare array-based and linked-list-based queue implementations. [5 Marks, L4]
- Q5. Design a queue-based solution for a printer scheduling system. [5 Marks, L5]